The Option Value of Rollback

Supplementary material: full proofs and verification protocol

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This supplement contains the complete proofs of the formal results stated in the paper, the exact statement of every assumption, and the protocol of the numerical checks that accompany each result. Numbering of theorems, propositions, definitions and assumptions matches the paper. Every number quoted here is generated by the artifact.

1 Model, notation and assumptions

Periods are $t = 0, \dots, T-1$. A single carrier has a latent taint bit $z_t \in \{0,1\}$ and an observed amount of derived work $D_t \in \mathbb{N}$ accumulated since the last verified checkpoint. Recovery costs $R(D) = r_0 + r_1 D$. In each period the defender chooses wait (W), recover (R: roll back to the checkpoint and replay clean work), or a contingent audit (A: pay c_a, observe the outcome, then choose W or R within the same period). Period t runs in five steps: (1) action, (2) growth of derived work by ξ_t, (3) activation of a tainted carrier that was not recovered, with probability λ, (4) otherwise a monitor output and a belief update, and (5) infection of a clean carrier with probability ρ_t.

Assumption 1 (A1, taint)**.** $\mathbb{P}(z_0 = 1) = \pi$. At the end of period t a clean carrier becomes tainted with probability $\rho_t \in [0, 1)$. Taint persists until a recovery, which sets $z = 0$.

Assumption 2 (A2, derived work)**.** $D_{t+1} = D_t + \xi_t$ unless a recovery occurs in period t, where the increments are i.i.d., nonnegative, integer valued with finite support and mean Δ, and independent of all other randomness. A chosen recovery in period t resets D before the period's growth, which gives $D_{t+1} = \xi_t$. A forced recovery after an activation (A4) resets it after the growth, which gives $D_{t+1} = 0$. No result depends on this convention.

Assumption 3 (A3, sensing)**.** The monitor emits $o_t \in \{0,1\}$ with $\mathbb{P}(o_t = 1 \mid z = 1) = \sigma_1$ and $\mathbb{P}(o_t = 1 \mid z = 0) = \sigma_0$. An audit is positive with probability d if $z = 1$ and α if $z = 0$. Given z all observations are independent. The monotone likelihood ratio $\sigma_1 \geq \sigma_0$ motivates the conjecture of Definition 3 but is not used by any proof.

Assumption 4 (A4, activation)**.** A tainted carrier that is not recovered in period t activates with probability $\lambda \in (0, 1)$. Activation is observed, costs ϕH (irreversible harm, $\phi \in [0, 1]$, $H \geq 0$) and triggers a mandatory recovery at cost $R(D_t + \xi_t)$, after which $z = 0$ and $D = 0$.

Assumption 5 (A5, costs and horizon)**.** $T < \infty$, all cost parameters are nonnegative, recovering a clean carrier costs an extra c_{fr}, and the terminal cost is $hD_T z_T$. The objective is the expected total cost.

Assumption 6 (A6)**.** $h \leq r_1$.

Assumption 7 (A7)**.** $d = 1$, $\alpha = 0$, $\rho_t \equiv 0$, $\sigma_0 = \sigma_1$, $c_{fr} = 0$, $\xi_t \equiv \Delta > 0$, and $R(D) \le hD$ for all $D \ge D_0$. A6 and A7 hold together only if $h = r_1$ and $r_0 = 0$.

Assumption 8 (A8, commitment game)**.** The defender commits to per-period audit probabilities $x \in X_B = \{x \in [0,1]^n : \sum_i x_i \le B\}$. The attacker observes x and poisons one carrier i at $t = 0$, succeeding with probability $s_i \in (0,1]$ at cost $\kappa_i \ge 0$, or abstains with payoff 0. Its payoff is $U_i(x_i) = s_i L_i(x_i, r_1) - \kappa_i$, where L_i is the defender's expected cost caused by a successful poison.

Assumption 9 (A9)**.** Carrier i has $D = 0$ when it is poisoned at $t = 0$, grows derived work deterministically by Δ_i per period and follows A4 and A5 with (λ_i, ϕ_i). In each period it is audited independently with probability x_i. An audit of a tainted carrier detects it with probability d and triggers recovery at cost $R(D_t)$. An undetected taint at T leaves residual harm hD_T. Moreover $h \ge r_1$, under which leaving a detected taint in place never saves more than r_0 (Section 4).

Assumption 10 (A10, regularity at r_1)**.** $\partial U_i/\partial x_i < 0$ on $[0, \min\{1, B\}]$ for every i. At a minimax commitment x^* with level $u^* = \max_i U_i(x_i^*)$ and attack set $S = \{i : U_i(x_i^*) = u^*\}$: $u^* > 0$, $0 < x_i^* < 1$ for $i \in S$, $\sum_i x_i^* = B$, and $U_j(0) < u^*$ for $j \notin S$. The condition $u^* > 0$ means that the attacker strictly prefers to attack. Without it every commitment with $\max_i U_i \le 0$ attains the value 0 of Definition 5, and the minimax commitment need not be unique.

Belief update. With $b = \mathbb{P}(z = 1 \mid \text{history})$, the posterior after no activation and monitor output o is $b^o = b(1-\lambda)\ell_1(o)/[b(1-\lambda)\ell_1(o) + (1-b)\ell_0(o)]$ with $\ell_z(1) = \sigma_z$ and $\ell_z(0) = 1 - \sigma_z$, followed by $b \mapsto b + (1-b)\rho_t$. An audit outcome updates b to $b^+ = bd/(bd + (1-b)\alpha)$ or $b^- = b(1-d)/(b(1-d) + (1-b)(1-\alpha))$.

Plans. From a decision point (t, D, b), a deterministic *plan* σ maps every finite sequence of future observations (monitor outputs, audit outcomes, activation indicators and work increments) to an action. Under A2 and A3 the set Σ_t of plans is finite and does not depend on (D, b). Write $J_\sigma(t, D, b)$ for its expected cost, $Q_a(t, D, b)$ for the minimum over plans whose first action is a, and $V_t = \min_a Q_a$. Optimal history-dependent policies are attained by plans because, for a fixed initial condition, D_s and b_s are functions of the observation history. The recovery region is $\mathcal{R}_t(D) = \{b : Q_{\mathsf{R}} \le \min(Q_{\mathsf{W}}, Q_{\mathsf{A}})\}$ and the audit region $\mathcal{A}_t(D)$ is the set where A attains the minimum.

2 Structure of the optimal adaptation policy

Lemma 1 (affine decomposition)**.** *Assume A1 to A5. For every plan $\sigma \in \Sigma_t$: (a) for fixed (t, D), $J_\sigma(t, D, b) = bJ_\sigma^1(t, D) + (1-b)J_\sigma^0(t, D)$, and (b) for fixed (t, b), $J_\sigma(t, D, b) = a_\sigma(t, b) + D\,\beta_\sigma(t, b)$ with $\beta_\sigma = r_1 \mathbb{P}_\sigma(E_{\text{rec}}) + h\mathbb{P}_\sigma(E_{\text{res}})$, where E_{rec} is the event that some recovery (chosen, after an audit, or forced) occurs in periods $t, \dots, T-1$, and E_{res} the event that none occurs and $z_T = 1$.*

Proof. (a) Under a fixed plan the actions are functions of the observations. Hence the conditional law of the future given z_t does not depend on b. Conditioning on z_t gives the mixture. (b) The law of the observation process under σ does not depend on D: increments are independent of everything (A2), activations and monitor outputs depend only on z, and actions are functions of observations. The initial D enters costs only through the first recovery, which costs $r_0 + r_1(D + \Xi)$ with Ξ the increments accumulated so far, and, if no recovery occurs, through the terminal cost $h(D + \Xi_T)z_T$. All later costs start from $D = 0$. Hence the realised cost equals $D\,[r_1 \mathbf{1}_{E_{\text{rec}}} + h\mathbf{1}_{E_{\text{res}}}]$ plus a term that does not depend on D. Taking expectations proves the claim. □

Theorem 1 (recovery structure, A1 to A5, A6 for (iii))**.** *(i) $V_t(D,\cdot)$ is concave and piecewise linear on $[0,1]$. (ii) $\mathcal{R}_t(D)$ is a closed interval, possibly empty. (iii) Under A6, $\mathcal{R}_t(D+k) \subseteq \mathcal{R}_t(D)$ for every integer $k \geq 1$. (iv) If $1 \in \mathcal{R}_t(D)$, then $\mathcal{R}_t(D) = [\theta_r(t,D), 1]$, where $\theta_r(t,D)$ denotes the left endpoint, and every belief at which auditing is strictly better than recovering lies below $\theta_r(t,D)$. If in addition A6 holds and $1 \in \mathcal{R}_t(D+k)$, then $\theta_r(t,D+k) \geq \theta_r(t,D)$. (v) If $d = 1$ and $\alpha = 0$, $\mathcal{A}_t(D)$ is a closed interval, possibly empty.*

Proof. (i) By Lemma 1a, $V_t(D,b) = \min_{\sigma\in\Sigma_t} J_\sigma(t,D,b)$ is a minimum of finitely many affine functions of b. This is the classical result of Smallwood and Sondik (1973). (ii) After a recovery in period t the carrier is clean and the belief is 0. The rest of the period has no activation, and the continuation is $\mathbb{E}_\xi V_{t+1}(\xi,\rho_t)$. Hence $Q_{\mathsf{R}}(t,D,b) = R(D) + (1-b)c_{fr} + \mathbb{E}_\xi V_{t+1}(\xi,\rho_t)$ is affine in b. The function $G(b) = \min(Q_{\mathsf{W}}, Q_{\mathsf{A}})(t,D,b)$ is a minimum over the plans whose first action is W or A. It is therefore concave and continuous. Then $\mathcal{R}_t(D) = \{b : G(b) - Q_{\mathsf{R}}(t,D,b) \geq 0\}$ is a superlevel set of a continuous concave function on $[0,1]$, which is a closed interval. This is the argument of Lovejoy (1987, Oper. Res. 35(4)) for the convexity of stopping regions. (iii) Fix (t,b) and let $\Sigma_t^{\neg\mathsf{R}}$ be the plans whose first action is W or A. By Lemma 1b, for $\sigma \in \Sigma_t^{\neg\mathsf{R}}$, $J_\sigma(t,D,b) = a_\sigma + D\beta_\sigma$ with $\beta_\sigma \leq r_1[\mathbb{P}(E_{\text{rec}}) + \mathbb{P}(E_{\text{res}})] \leq r_1$ under A6, because the two events are disjoint. Also $Q_{\mathsf{R}}(t,D,b) = r_1 D + c(t,b)$ with c independent of D. Therefore

$$F(D) = \min(Q_{\mathsf{W}}, Q_{\mathsf{A}})(t,D,b) - Q_{\mathsf{R}}(t,D,b) = \min_{\sigma\in\Sigma_t^{\neg\mathsf{R}}} \left[a_\sigma - c + D(\beta_\sigma - r_1)\right]$$

is a minimum of nonincreasing functions of D and is nonincreasing. If $b \in \mathcal{R}_t(D+k)$ then $F(D+k) \geq 0$, which gives $F(D) \geq 0$ and $b \in \mathcal{R}_t(D)$. (iv) A closed interval in $[0,1]$ that contains 1 has the form $[\theta,1]$. A belief at which auditing is strictly better than recovering is not in $\mathcal{R}_t(D)$ and is therefore below θ_r. If A6 holds and $1 \in \mathcal{R}_t(D+k)$, then by (iii) $[\theta_r(t,D+k),1] \subseteq \mathcal{R}_t(D) = [\theta_r(t,D),1]$. (v) With $d = 1$ and $\alpha = 0$ an audit reveals z: the posterior is 1 with probability b and 0 with probability $1-b$. Hence $Q_{\mathsf{A}}(t,D,b) = c_a + b\,M(1) + (1-b)\,M(0)$ with $M(\beta) = \min(Q_{\mathsf{W}}, Q_{\mathsf{R}})(t,D,\beta)$, which is affine in b. The function $\min(Q_{\mathsf{W}}, Q_{\mathsf{R}})$ is concave by the argument of (ii). Hence $\mathcal{A}_t(D) = \{b : \min(Q_{\mathsf{W}}, Q_{\mathsf{R}})(t,D,b) - Q_{\mathsf{A}}(t,D,b) \geq 0\}$ is a closed interval. □

Relation to classical results. Parts (i) and (ii) are the concavity of POMDP values and the convexity of stopping sets. The reset structure is that of partially observed machine replacement (Krishnamurthy 2016), which Hammar and Stadler (DSN 2024) use to prove threshold recovery strategies for intrusion tolerance. Part (iii) concerns monotonicity in the *observed* derived work. It uses the linearity of replay cost in D, and its direction is the opposite of the conjecture of Definition 3.

Proposition 2 (the two-threshold structure fails, A1 to A5)**.** *Each instance has $T = 2$, $\pi = 0.5$, deterministic growth $\xi_t \equiv 1$, $\rho_t \equiv 0$ and $c_{fr} = 0$. (a) Perfect audits, $\lambda = 0.2$, $\phi = 0$, $H = 10$, $h = 4$, $r_0 = 4$, $r_1 = 1$, $c_a = 2$, $\sigma_0 = 0.2$, $\sigma_1 = 0.9$, $D_0 = 10$: at $(t,D) = (0,10)$ the optimal first action is W at $b = 0.20$, A at 0.55, W at 0.82 and R at 0.95. (b) A6 holds, $\lambda = 0.3$, $\phi = 1$, $H = 20$, $h = 1$, $r_0 = 1$, $r_1 = 1$, $c_a = 2$, $d = 0.9$, $\alpha = 0.05$, $\sigma_0 = 0.1$, $\sigma_1 = 0.4$: the recovery threshold at $t = 0$ is 0.213 for $D = 2$ and 0.597 for $D = 6$. At $b = 0.4$ recovery is optimal for $D = 2$ and an audit for $D = 6$. (c) Imperfect audits, $\lambda = 0.2$, $\phi = 0.5$, $H = 2$, $h = 4$, $r_0 = 2$, $r_1 = 2$, $c_a = 0.25$, $d = 0.8$, $\alpha = 0.3$, $\sigma_0 = 0.2$, $\sigma_1 = 0.7$, $D_0 = 6$: at $(t,D) = (0,6)$ the optimal first action is $\mathsf{W}, \mathsf{A}, \mathsf{W}, \mathsf{A}, \mathsf{R}$ at $b = 0.05, 0.18, 0.33, 0.6, 0.95$.*

Proof. For each listed belief, the expected cost of every deterministic two-period plan was computed by exhaustive enumeration (18,396 plans in (a), 63,756 in (b) and (c)), and the minimum over plans

with each first action agrees with exact backward induction to 10^{-9} (Table 1). The thresholds in (b) are exact. Every plan's cost is affine in b, and its two coefficients follow from its costs at two interior beliefs. The threshold $\theta_r(0, D)$ is the smallest b at which the lower envelope of the 63,756 plans that recover first meets the lower envelope of the others, located by bisection on these exact envelopes to 10^{-12}. The values are 0.2130 for $D = 2$ and 0.5968 for $D = 6$. A narrow audit band lies just below the threshold for $D = 2$. In (a) the wait region contains $[0, 0.383]$ and $[0.758, 0.890]$ with audits in between, and no thresholds $\theta_a \leq \theta_r$ exist. By Theorem 1(v) the audit region in (a) is an interval, and (c) shows that with imperfect audits it need not be. In (b) the threshold rises with D, against the conjecture that it falls. □

Mechanism of (a). At $b = 0.82$ the free monitor output of the next period resolves most of the uncertainty: an alarm raises the belief to 0.94, where recovery is clearly optimal, and no alarm lowers it to 0.31, where one audit is bought. Waiting saves the audit cost in the alarm branch at the price of one more unit of replay. At $b = 0.55$ an alarm only reaches 0.81, which still needs an audit, and the audit is bought at once. Ross (1971) observed the same failure of the intuitive three-region structure in a two-state inspection problem.

Table 1: Exhaustive enumeration for Proposition 2: minimum expected cost over all deterministic plans by first action.

instance	belief b	plans	best W	best R	best A	optimal
(a)	0.20	18,396	4.034	14.000	4.800	W
(a)	0.55	18,396	10.030	14.000	9.700	A
(a)	0.82	18,396	13.259	14.000	13.480	W
(a)	0.95	18,396	14.632	14.000	15.300	R
(b), $D = 2$	0.40	63,756	5.298	3.000	3.746	R
(b), $D = 6$	0.40	63,756	7.097	7.000	5.466	A
(c)	0.05	63,756	1.346	14.000	1.596	W
(c)	0.18	63,756	4.618	14.000	4.500	A
(c)	0.33	63,756	7.625	14.000	7.789	W
(c)	0.60	63,756	12.930	14.000	11.460	A
(c)	0.95	63,756	16.190	14.000	14.250	R

Search statistics. Over 1,000 random instances ($T \in \{3, \ldots, 8\}$, 36,247 cells (t, D), a grid of 1,001 beliefs), the solver found no violation of Theorem 1(ii), no violation of (iii) among 7,166 comparisons in instances satisfying A6, and no violation of (v) in 400 random perfect-audit instances. The two-threshold order was broken in 39 instances, the audit region was disconnected in 16, the recovery threshold increased with D in 556 and the first non-wait threshold in 571. Exact recursion confirmed 41 of the 41 flagged cells it re-checked: the first 25 of the order-break instances and all 16 instances with a disconnected audit region. The increasing thresholds were not re-checked exactly.

3 Exact audit timing in the perfect-audit submodel

Under A7 write $\pi = \pi_0$, $D_s = D_0 + \Delta s$, $q_\tau = \pi(1-\lambda)^\tau + (1-\pi)$ (the probability of no activation before period τ), $N_T = \sum_{s=0}^{T-1}(1-\lambda)^s = (1-(1-\lambda)^T)/\lambda$ and $g_0 = r_1\Delta + \lambda\phi H$.

Lemma 2 (reduction to first-intervention policies, A1 to A5 and A7)**.** *There is an optimal policy of the following form: wait until a first intervention time $\tau \in \{0, \dots, T-1\}$ (if no activation occurred before τ) and then either audit and recover on a positive outcome or recover blind, or never intervene.*

Proof. Since $\sigma_0 = \sigma_1$, monitor outputs are independent of the taint and of activations. A policy that conditions its first intervention on them induces a probability mixture over first-intervention times, and its expected cost is the same mixture of the costs of deterministic first-intervention times. It is therefore no better than the best of them. Before the first intervention the only informative event is the absence of activation. After an activation the mandatory recovery leaves a clean carrier that stays clean ($\rho_t \equiv 0$), and the remaining cost is 0 under W. After a blind recovery the same holds. After a negative audit the carrier is known clean ($d = 1$) and the remaining cost is 0. After a positive audit the carrier is known tainted ($\alpha = 0$). Recovering at once costs $R(D_\tau)$. Any other continuation either recovers later at cost $R(D_s) \geq R(D_\tau)$, possibly after paying ϕH, or never recovers and pays $hD_T \geq R(D_T) \geq R(D_\tau)$ by A7 and the monotonicity of R. Immediate recovery is therefore optimal. □

Under Lemma 2 the candidate costs are

$$J_A(\tau) = \pi\Big[\sum_{s=0}^{\tau-1} \lambda(1-\lambda)^s\big(\phi H + R(D_{s+1})\big) + (1-\lambda)^\tau R(D_\tau)\Big] + c_a q_\tau,$$
$$J_B(\tau) = \pi \sum_{s=0}^{\tau-1} \lambda(1-\lambda)^s\big(\phi H + R(D_{s+1})\big) + q_\tau R(D_\tau),$$
$$J_N = \pi\Big[\sum_{s=0}^{T-1} \lambda(1-\lambda)^s\big(\phi H + R(D_{s+1})\big) + (1-\lambda)^T hD_T\Big].$$

Theorem 3 (exact audit timing, A1 to A5 and A7)**.** *(i) For $0 \leq \tau \leq T-2$, $J_A(\tau+1) - J_A(\tau) = \pi(1-\lambda)^\tau g$ with $g = r_1\Delta + \lambda(\phi H - c_a)$. (ii) An optimal policy is one of four corners: audit at 0, blind recovery at 0, audit at $T-1$, never intervene. If $g > 0$ the best audit time is 0, and if $g < 0$ it is $T-1$. (iii) Let $\pi > 0$ and $T \geq 2$. Auditing at 0 is at least as good as every later audit for all $r_1 \geq 0$ if and only if $\phi H \geq c_a$. If $\phi H < c_a$, it is strictly better than every later audit exactly when $r_1 > r_1^\star = \lambda(c_a - \phi H)/\Delta$, and the two tie at $r_1 = r_1^\star$. As $r_1 \to 0$, $g \to \lambda(\phi H - c_a)$. (iv) The myopic rule of Definition 4 audits at $t = 0$ if $\pi g_0 \geq c_a$ and never otherwise. If it audits, $J_{\text{my}} - J^\star = \max\{0, c_a - (1-\pi)R(D_0)\} \leq c_a$. If it defers,*

$$0 \leq J_{\text{my}} - J^\star < c_a N_T + \pi(1-\lambda)^T\big(hD_T - R(D_T)\big), \qquad N_T \leq \min\{T, 1/\lambda\},$$

and the supremum, over all instances satisfying A7 in which the rule defers, of the ratio of the gap to the right-hand side equals 1.

Proof. (i) Subtracting, the terms with $s < \tau$ cancel and

$$\begin{aligned} J_A(\tau+1) - J_A(\tau) &= \pi(1-\lambda)^\tau\big[\lambda(\phi H + R(D_{\tau+1})) + (1-\lambda)R(D_{\tau+1}) - R(D_\tau)\big] + c_a(q_{\tau+1} - q_\tau) \\ &= \pi(1-\lambda)^\tau\big[\lambda\phi H + r_1\Delta\big] - c_a\pi\lambda(1-\lambda)^\tau, \end{aligned}$$

using $R(D_{\tau+1}) - R(D_\tau) = r_1\Delta$ and $q_{\tau+1} - q_\tau = -\pi\lambda(1-\lambda)^\tau$. (ii) By (i) the increments of J_A have the sign of g for every τ, and the minimum over τ is at $\tau = 0$ if $g > 0$ and at $\tau = T-1$ if $g < 0$ (any τ if $g = 0$). For blind recovery, $J_B(\tau+1) - J_B(\tau) = \pi\lambda(1-\lambda)^\tau\phi H + q_\tau r_1\Delta \geq 0$. Hence

$\min_\tau J_B(\tau) = J_B(0) = R(D_0)$. With Lemma 2 the optimum is the minimum of $J_A(0)$, $J_A(T-1)$, $J_B(0)$ and J_N. (iii) With $\pi > 0$ and $T \geq 2$ there is at least one later audit time and the increments of (i) have the sign of g. Hence auditing at 0 is at least as good as every later audit iff $g \geq 0$, and strictly better iff $g > 0$. For all $r_1 \geq 0$, $g \geq 0$ iff $\lambda(\phi H - c_a) \geq 0$. If $\phi H < c_a$, $g > 0$ iff $r_1 > r_1^\star$ (using $\Delta > 0$), with equality at $r_1^\star$. (iv) Before any intervention no activation has been observed, and $b_t = \pi(1-\lambda)^t/q_t$ decreases in t. With $d = 1$ and $\alpha = 0$ the rule audits at t if $b_t g_0 \geq c_a$, hence at 0 if $\pi g_0 \geq c_a$ and never otherwise. After a positive audit $b = 1$ and its recovery test $g_0 \geq 0$ holds. Thus $J_{\text{my}} \in \{J_A(0), J_N\}$. Telescoping $\sum_{s<T} \lambda(1-\lambda)^s R(D_{s+1}) + (1-\lambda)^T R(D_T) - R(D_0) = r_1 \Delta N_T$ and $\lambda N_T = 1 - (1-\lambda)^T$ give

$$J_N - J_A(0) = \pi g_0 N_T + \pi(1-\lambda)^T \big(hD_T - R(D_T)\big) - c_a.$$

Next, $J_A(0) - J^\star \leq c_a$ in every case. First, $J_A(0) - J_B(0) = c_a - (1-\pi)R(D_0) \leq c_a$. Second, $J_A(0) - J_A(T-1) = \pi(\lambda c_a - g_0)N_{T-1} \leq \pi c_a \lambda N_{T-1} \leq c_a$. Third, $J_A(0) - J_N = c_a - \pi g_0 N_T - \pi(1-\lambda)^T(hD_T - R(D_T)) \leq c_a$, because $g_0 \geq 0$ and A7 gives $hD_T \geq R(D_T)$. If the rule defers, $\pi g_0 < c_a$ and $J_{\text{my}} - J^\star = (J_N - J_A(0)) + (J_A(0) - J^\star) < c_a(N_T - 1) + \pi(1-\lambda)^T(hD_T - R(D_T)) + c_a$. If it audits, $\pi g_0 \geq c_a$, and the identity with $N_T \geq 1$ and A7 gives $J_N \geq J_A(0)$. Also $g_0 \geq c_a/\pi \geq \lambda c_a$, which gives $J_A(0) \leq J_A(T-1)$ and $J^\star = \min\{J_A(0), J_B(0)\}$. *Supremum.* Take $r_1 = 0$, $\phi = 1$, $D_0 = 1$, $\pi\lambda H = (1-\epsilon)c_a$, any $r_0 > 0$ and $h = r_0$. These choices satisfy A7. The rule defers. Then $J^\star \leq J_B(0) = r_0$ and $J_{\text{my}} = J_N \geq \pi\phi H(1-(1-\lambda)^T) = (1-\epsilon)c_a N_T$. The gap is therefore at least $(1-\epsilon)c_a N_T - r_0$, while the right-hand side equals $c_a N_T + \pi(1-\lambda)^T r_0 \Delta T$. As $r_0 \to 0$ the ratio tends to at least $1-\epsilon$, and ϵ is arbitrary. The ratio never exceeds 1 by the bound itself. *Unbounded residual term.* Take $\pi = 1$, $r_1 = 0$, $\phi = 1$, $\lambda H = (1-\epsilon)c_a$ and any h with $r_0 \leq hD_0$, which satisfies A7. The rule defers, $J^\star \leq J_B(0) = r_0$ and $J_N = (1-\epsilon)c_a N_T + (1-(1-\lambda)^T)r_0 + (1-\lambda)^T hD_T$. The gap is at least $(1-\epsilon)c_a N_T + (1-\lambda)^T(hD_T - R(D_T))$, which grows without bound in h, and its ratio to the right-hand side of (iv) is at least $1-\epsilon$ for every h. □

Relation to classical results. Part (iii) is an analogue of the irreversibility effect of Arrow and Fisher (1974) and Henry (1974): deferring an audit keeps the option to recover, and the option is worthless for the irreversible fraction ϕ of the harm. No quasi-option-value construction is involved. The term $-\lambda c_a$ is the continuation value the myopic rule drops: activation itself reveals the taint for free.

Numerical check. Over 600 random instances satisfying A7 the corner formula matched exact backward induction with maximal absolute error 3×10^{-15}, the best audit time was always the corner predicted by the sign of g (0 violations), the myopic rule deferred in 339 instances, its exact value matched the closed form in 600 instances, and the bound of (iv) was never violated (0 violations). In the family of the supremum argument with $\epsilon = 0.05$ the ratio of gap to bound is 0.950 for $r_0 = 0.01$.

4 Attacker shift and coverage reallocation

Under A8 and A9 let $q_i(x) = (1-dx)(1-\lambda_i)$ and $N_i(x) = \sum_{t=0}^{T-1} q_i(x)^t$. A9 makes recovery after a detection mandatory, as A4 does after an activation. The condition $h \geq r_1$ bounds the price of this choice: a detected taint left in place is later recovered at $R(D_s) \geq R(D_t)$ (after a detection or an activation) or reaches T and costs $hD_T \geq r_1 D_T \geq r_1 D_t = R(D_t) - r_0$. Leaving it therefore never saves more than the fixed price r_0.

Theorem 4 (attacker shift and reallocation, A8, A9, and A10 for (iii))**.** *(i)* $L_i(x, r_1) = F_i(x) + r_1 E_i(x)$ *with*

$$F_i(x) = (1 - dx)\lambda_i \phi_i H\, N_i(x) + r_0\big(1 - q_i(x)^T\big) + h\Delta_i T\, q_i(x)^T,$$
$$E_i(x) = \textstyle\sum_{t=0}^{T-1} q_i(x)^t \big[dx\, \Delta_i t + (1 - dx)\lambda_i \Delta_i (t+1)\big].$$

Both are nonnegative polynomials in x*. At* $x = 0$ *the irreversible term of* F_i *equals* $\phi_i H(1 - (1 - \lambda_i)^T) = \lambda_i \phi_i H\, T\,(1 + O(\lambda_i T))$*. (ii) For any fixed commitment* x*, if carrier* i *is a best response at* r_1 *and* j *at* $r_1' > r_1$ *(abstention counts as a carrier with* $sE = 0$*), then* $s_j E_j(x_j) \geq s_i E_i(x_i)$*. At* $r_1 = 0$ *the best response maximises* $s_i F_i(x_i) - \kappa_i$ *if that maximum is positive, and abstains otherwise. (iii) Under A10 the minimax commitment is unique and differentiable at* r_1*, and*

$$\frac{dx_i^*}{dr_1} = \frac{s_i E_i(x_i^*) - \bar{E}}{|\partial U_i / \partial x_i|}\ (i \in S), \qquad \frac{du^*}{dr_1} = \bar{E} = \frac{\sum_{j \in S} w_j s_j E_j(x_j^*)}{\sum_{j \in S} w_j} \geq 0,$$

with $w_j = 1/|\partial U_j / \partial x_j|$*, and* $dx_j^*/dr_1 = 0$ *for* $j \notin S$*.*

Proof. (i) Condition on the first event. In period t the carrier is still tainted with probability q^t. It is then detected with probability dx (cost $R(\Delta_i t)$), or else activates with probability λ_i (cost $\phi_i H + R(\Delta_i(t+1))$). With probability q^T it survives to T (cost $h\Delta_i T$). Summing and separating the coefficient of r_1 gives $L_i = F_i + r_1 E_i$. The r_0 terms sum to $r_0 \sum_t q^t(1-q) = r_0(1 - q^T)$ because $dx + (1 - dx)\lambda_i = 1 - q$. At $x = 0$, $\lambda_i N_i(0) = 1 - (1 - \lambda_i)^T$, and the expansion follows from the binomial series. (ii) For fixed x, $U_k(r_1) = [s_k F_k(x_k) - \kappa_k] + r_1 s_k E_k(x_k)$ is affine in r_1 with slope $\beta_k = s_k E_k(x_k)$. From $U_i(r_1) \geq U_j(r_1)$ and $U_j(r_1') \geq U_i(r_1')$, adding gives $(\beta_j - \beta_i)(r_1' - r_1) \geq 0$. (iii) *Smoothness.* $U_i(x, r_1) = s_i(F_i(x) + r_1 E_i(x)) - \kappa_i$ is a polynomial in x and affine in r_1, hence jointly C^∞. Every feasible coverage lies in $[0, \min\{1, B\}]$, where A10 and compactness give $\partial_x U_i \leq -\delta < 0$. For r_1' near r_1 the inequality $\partial_x U_i(\cdot, r_1') < 0$ persists because $\partial_x E_i$ is bounded. *Water filling.* Because $u^* > 0$, the objective $\max\{0, \max_i U_i\}$ equals $\max_i U_i$ near x^*, and lowering the maximum payoff lowers the objective. At a minimax commitment every carrier with positive coverage attains the maximum. Otherwise move a small amount ε of its coverage, in equal parts, to the carriers that attain the maximum. Their coverage stays in $[0, \min\{1, B\}]$ for small ε (it is below 1 under A10 and the total is unchanged), and each of their payoffs falls strictly while the donor stays below the maximum, which contradicts optimality. In particular $x_j^* = 0$ for $j \notin S$. *Uniqueness.* Let x' be any minimax commitment. For $j \notin S$, $U_j(x_j') \leq U_j(0) < u^*$. For $i \in S$, $U_i(x_i') \leq u^* = U_i(x_i^*)$ gives $x_i' \geq x_i^*$, and $\sum_{i \in S} x_i' \leq B = \sum_{i \in S} x_i^*$ forces $x' = x^*$. This is the equalisation of ORIGAMI (Kiekintveld et al. 2009). *Local constancy of* S*.* The value $v(r_1) = \min_{x \in X_B} \max_i U_i(x_i, r_1)$ is continuous, it stays positive near r_1, and the solution correspondence is upper hemicontinuous (Berge's maximum theorem, continuous objective, fixed compact feasible set). For r_1' near r_1 every minimax commitment $x(r_1')$ is therefore close to x^*. For $j \notin S$, $U_j(0, r_1') < v(r_1')$ by continuity. Then j does not attain the maximum and water filling gives $x_j(r_1') = 0$. For $i \in S$, $x_i(r_1')$ is close to $x_i^* \in (0, 1)$ and positive, and i attains the maximum. The budget binds at r_1', since otherwise raising the coverage of the maximisers would lower the maximum. *Implicit function.* Near (x_S^*, u^*, r_1) the system $U_i(x_i, r_1) - u = 0$ for $i \in S$ and $\sum_{i \in S} x_i - B = 0$ has a Jacobian with respect to (x_S, u) with determinant $\pm \prod_{i \in S} \partial_x U_i \cdot \sum_{i \in S} 1/\partial_x U_i \neq 0$. The implicit function theorem gives a unique C^1 solution near r_1, which by local constancy and uniqueness is the minimax commitment. Differentiating $U_i(x_i^*(r_1), r_1) = u^*(r_1)$ gives $\partial_x U_i\, x_i^{*\prime} + s_i E_i = u^{*\prime}$. Summing $x_i^{*\prime} = (s_i E_i - u^{*\prime})/|\partial_x U_i|$ over S and using $\sum_i x_i^{*\prime} = 0$ yields $u^{*\prime} = \bar{E} \geq 0$. □

Relation to the classical result. ORIGAMI and the analysis of Korzhyk et al. (2011) equalise attacker utilities over the attack set for fixed target values. Here the values depend on the recovery infrastructure, and part (iii) gives the first-order shift of the minimax coverage when the price of rollback changes. Its sign is set by the reversible exposure $s_i E_i$ relative to the weighted mean $\bar{E}$. The irreversible fraction ϕ_i enters only through F_i, which sets x^* and with it $E_i(x_i^*)$, the weights and $\bar{E}$. It can change the sign only through these quantities. The same structure appears in inspection games (Avenhaus, von Stengel and Zamir 2002) with fixed values.

Proposition 5 (no simple migration to $\arg\max \phi$, A8 and A9)**.** *(a) In the four-carrier instance of the paper ($h = 10$) at $r_1 = 0$ without coverage, the best response is the memory note with $\phi = 0$ (payoff 65.76), and the queued call with $\phi = 1$ yields the least (13.76). (b) The carrier with the largest ϕ can lose coverage when rollback gets cheaper. (c) Where the attack set changes, the minimax coverage need not be differentiable.*

All payoffs of (a) use $T = 20$, $H = 20$, $d = 0.9$ and $r_0 = 3$. With $h = 1$, the residual price of E1 and E3, the best response without coverage is the skill (payoff 9.24) and the queued call still yields the least (5.80). The point of (a), that the attacker does not go to the largest ϕ, therefore holds at both residual prices.

Proof. (a) is read from the closed forms of Theorem 4(i). (b) Take carriers $a = (\Delta, \lambda, \phi, s, \kappa) = (1, 0.02, 1, 0.5, 1)$ and $b = (0.1, 0.2, 0, 0.8, 0.5)$ with $T = 20$, $H = 20$, $h = 2$, $r_0 = 1$, $d = 0.9$ and $B = 0.5$, for which A9 holds when $r_1 \in [0, 2]$. On a grid of 21 values of $r_1 \in [0, 2]$ both carriers stay in the attack set and A10's derivative condition holds. The coverage of a falls from 0.48 at $r_1 = 2$ to 0.24 at $r_1 = 0$. Its reversible exposure $s_a E_a$ is 1.66 against 0.16 for b at $r_1 = 0$, a ratio of 10.7, and Theorem 4(iii) moves coverage away from it as r_1 falls. In addition, among the 1,187 random games that satisfy A9 at $r_1 = 2$ ($h \geq 2$) and have a positive attacker level at $r_1 = 0$ and $r_1 = 2$, the largest-ϕ carrier received less coverage at $r_1 = 0$ than at $r_1 = 2$ in 392. (c) In the four-carrier instance the queued call leaves the attack set at $r_1 \approx 7.27$ (bisection), where the one-sided finite-difference derivatives of its coverage are -0.0076 and 0.0. □

Numerical check. The closed form of (i) agreed with Monte Carlo within 3.5 standard errors in 80 of 80 cells. Best-response slopes were monotone in r_1 in all 1,500 random games (0 violations), a check of the implementation of an algebraic identity. Under A10, checked on a 401-point coverage grid, the derivative formula matched central finite differences within 5% in 352 of 352 games with matching signs in 352. 40 random games failed the other conditions of A10 and 8 more had a nonpositive level, and all were excluded. In the four-carrier instance A10 holds at every grid value of r_1 (the queued call enters the attack set between grid values, near $r_1 \approx 7.27$).

5 Accuracy of the belief grid

The planner tables use linear interpolation on a uniform belief grid. Against the exact recursion over the reachable belief tree, on the E1 base instance with $T \leq 6$, every derived work $D \leq D_0$ and 37 beliefs (most of them off the grid), the relative error of the 1,001-point grid is below 5×10^{-5}. Longer horizons are out of reach of the exact recursion.

At the horizon of E3 ($T = 24$) we compare every ROVER table (four carriers, $r_1 \in \{0, 0.1, 1, 10\}$) with an 8,001-point grid over all periods, every derived work $D \leq t$ and 1,001 beliefs. The 501-point tables differ in value by at most 0.0028 (0.15% relative) and in the chosen action in at most 0.002% of these states, and the errors fall as the grid is refined. With 2,001-point tables ROVER takes the

same decisions, with the same losses, in all 960 held-out main episodes at the four replay prices. This is a convergence study against a finer grid and not an error bound.

6 Sensitivity grid of E1

Table 2 varies one parameter of the E1 base instance at a time and reports the optimal audits per 100 periods. The pattern of the main text (with $\phi = 1$ audits are worth more than with $\phi = 0$ when rollback is free, and with $\phi = 0$ the audit rate rises with r_1) holds in 9 of the 13 distinct configurations. It breaks for $c_a = 2$, $d = 0.5$, $d = 0.7$ and $\lambda = 0.02$. With dear or weak audits the $\phi = 1$ policy replaces audits by precautionary recoveries when rollback is free, and with rare activations neither policy intervenes at $r_1 = 10$, where a recovery costs more than the expected harm.

Table 2: E1 sensitivity grid: optimal audits per 100 periods for $\phi \in \{0, 1\}$ and $r_1 \in \{0, 1, 10\}$, one parameter changed at a time from the base instance.

	$\phi = 0$, $r_1 = 0, 1, 10$			$\phi = 1$, $r_1 = 0, 1, 10$			pattern
$c_a = 0.25$	4.5	5.0	17.6	17.9	14.5	23.6	holds
$c_a = 0.5$	3.5	2.7	8.9	15.7	11.8	14.0	holds
$c_a = 1$	1.6	1.9	8.1	5.3	7.7	9.5	holds
$c_a = 2$	0.0	0.0	5.6	0.0	3.6	7.1	breaks
$d = 0.5$	0.0	0.0	7.2	0.0	3.8	8.1	breaks
$d = 0.7$	0.7	0.8	7.3	0.4	5.3	8.8	breaks
$d = 0.9$	1.6	1.9	8.0	5.3	7.7	9.5	holds
$d = 1$	2.0	1.9	8.0	6.7	7.6	8.7	holds
$\lambda = 0.02$	2.0	3.1	0.0	2.2	4.1	0.0	breaks
$\lambda = 0.05$	1.9	2.5	5.7	2.7	4.6	6.9	holds
$\lambda = 0.1$	1.6	1.9	8.0	5.3	7.7	9.4	holds
$\lambda = 0.2$	1.1	0.7	9.1	11.5	8.1	11.1	holds
$\sigma_1 = 0.1$	4.0	0.0	10.8	13.4	10.5	11.2	holds
$\sigma_1 = 0.25$	3.2	1.4	8.0	12.0	8.9	11.8	holds
$\sigma_1 = 0.4$	1.6	1.9	8.1	5.3	7.7	9.5	holds
$\sigma_1 = 0.7$	0.7	0.3	6.0	1.4	4.1	6.8	holds

7 Development history and mechanics of E3

On the development streams, our first full run used a constant insertion hazard and a replay-only executor. It lost to rollback only at $r_1 = 0$ (5.70 against 5.28) and to reset all at $r_1 = 10$ (42.43 against 28.39). Both changes are kept as ablations, and on held-out data each original loses at its extreme: without the window at $r_1 = 0$ (5.90 against 4.85, Holm $p = 0.001$) and with replay only at $r_1 = 10$ (39.64 against 18.82).

The held-out run issued 1,669,886 git commands. A recovery took 1542 ms of git time on average and re-executed or cherry-picked the work since the checkpoint (54,298 re-executions, 22,569 cherry-picks, 4,448 conflicts). Irreversible pushes delivered 10,016 canary lines to the bare remote, and no policy could remove them.

8 Exploratory analyses added after the confirmatory run

The analyses below were run after the confirmatory E3 run and are not part of its protocol. Their definitions are our choices and are stated here in full.

One-step controllers in E1. The myopic rule (Definition 4) waits or audits and never recovers blindly. A one-step controller chooses $\arg\min\{Q_W, Q_R, Q_A\}$ with the one-period transitions of Definition 1 and a fixed continuation value V_{t+1} in place of the optimal cost-to-go, and after an audit it recovers when its own $Q_R \leq Q_W$ at the posterior. With the terminal cost hDb as continuation it is a plain one-step lookahead. With the exact cost-to-go of never auditing it is one policy-improvement step on that policy (a rollout). Each is run with and without blind recovery and evaluated exactly on the belief grid as the myopic rule is, with a Monte Carlo cross-check (largest $|z| = 2.37$). Table 3 gives the gaps to the optimum on the E1 grid. Allowing blind recovery helps the rollout controller in 18 and hurts it in 25 of the 108 configurations of the E1 grid and the sensitivity grid, and the rollout controller beats the myopic rule in 101 of them.

Table 3: E1, gap to the optimal loss in % for the myopic rule, the myopic rule with the DP's recovery decisions, the one-step lookahead with blind recovery, and the rollout controller without and with blind recovery.

ϕ	r_1	DP loss	Myopic	Myopic, DP rec.	One-step	Rollout W/A	Rollout W/R/A
0	0	2.79	69.6	69.6	61.8	40.4	42.5
0	0.1	3.60	44.9	44.9	35.0	25.5	24.8
0	1	9.15	5.0	5.0	5.0	0.8	0.8
0	10	47.00	17.7	5.1	13.7	0.7	0.7
0.5	0	5.36	56.1	56.1	10.2	22.2	25.4
0.5	0.1	6.08	45.5	45.5	6.4	17.4	17.4
0.5	1	11.12	6.5	6.4	19.0	4.9	4.9
0.5	10	48.73	18.0	6.2	17.1	0.9	0.9
1	0	7.05	24.2	23.9	2.4	7.1	10.5
1	0.1	7.70	19.7	19.7	1.6	5.8	5.8
1	1	12.68	7.2	7.2	28.3	2.8	2.8
1	10	50.38	17.7	6.5	20.4	1.0	1.0

Publication harm in E3. The main runs have no benign releases. We add releases at rate $\nu \in \{1/24, 1/4\}$ per milestone, with no gate, to the main condition at $r_1 = 1$ for nine policies on the held-out cells (4,320 episodes). A release publishes the work branch to the bare remote. Every decision count and loss equals the corresponding main episode, and the leaks can therefore be priced after the fact. We count the distinct canary lines that releases publish and that no irreversible activation published before, and charge either $h_{\text{pub}} \in \{1, 2, 5, 20\}$ per such line or $H_{\text{rel}} \in \{1, 5, 20\}$ per release that contains one. Table 4 lists the leaks and the losses at the highest prices. With rare releases ROVER has the lowest loss at every price. With frequent releases it has the lowest loss up to a price of five per line or per release and drops to rank 3 at a price of 20, behind the immutable validator and periodic review. The testbed's counter of canary occurrences over remote refs exceeds the number of distinct leaked lines up to 3.1-fold (2.7-fold for ROVER) with frequent releases.

Table 4: E3 with releases: distinct newly leaked canary lines per episode, share of episodes with a leak, and mean loss with frequent releases without a publication price, at 20 per leaked line and at 20 per contaminated release.

	$\nu = 1/24$		$\nu = 1/4$		loss, $\nu = 1/4$		
Policy	lines	leak %	lines	leak %	none	20/line	20/release
ROVER	0.26	8	1.02	30	9.28	29.70	24.61
ROVER, no audits	0.55	15	1.58	39	10.37	41.95	33.37
Rollback only	0.33	12	1.22	38	11.36	35.77	30.61
No audit	0.70	20	2.00	46	11.86	51.86	45.20
Random audit	0.69	19	1.85	45	13.46	50.46	45.29
Myopic rule	0.67	18	1.83	45	11.50	48.08	40.66
Periodic review	0.06	4	0.34	22	16.07	22.82	23.07
Immutable validator	0.02	2	0.04	5	15.19	15.94	16.86
Reset all	0.30	10	0.98	34	25.94	45.60	40.77

Cost semantics in E3. An instrumented re-run of ROVER and tuned rollback only at $r_1 \in \{0.1, 1\}$ (960 episodes, decisions equal to the main runs) logs the planner's price $r_0 + \min(r_1, w)D$ at every recovery and the executed price. Executed recoveries cost at most 1.8% more on average, because cherry-pick conflicts force 0.12 to 0.21 extra re-executions per recovery. For carriers still tainted at the end, the surviving canary lines fall 12 to 28% short of the terminal prediction $hP_{\text{infl}}D$, because D counts dependent commits made before the insertion. Measured against hP_{infl} times the direct reads after the insertion, the ratio is 0.94 to 1.09. Transitive dependencies make D grow 8 to 22% faster than the planner's growth model for the carriers that are not read at every milestone.

Estimands. Next to the signed-rank test we test the mean of the paired per-stream differences with a paired t-test and a sign-flip permutation test (200,000 sign vectors). The primary decisions do not change (Holm $p < 0.001$ and $p < 0.001$ for H1, $p < 0.001$ and $p < 0.001$ for H2). Over the 41 comparisons against ROVER, 1 Holm decision changes. A_{12} as reported compares every stream of one policy with every stream of the other (0.63 for H1), while the paired share of streams on which ROVER is lower is 0.87.

Adaptive attacker. The attacker's payoff per carrier rests on 30 to 45 attacked episodes, and the reported maximum over four carriers is biased upward. Choosing the carrier on seeds 8 to 11 and scoring it on seeds 12 to 15, and the reverse, gives 8.5 against ROVER and 9.8 against the immutable validator.

Exact check of A10. For the E2 game and the instance of Proposition 5, F_i and E_i are polynomials in x_i (degree 20). We build them with exact rational coefficients from the stated parameters and count the real roots of $\partial U_i/\partial x_i = s_i(F_i' + r_1 E_i')$ on $[0, \min\{1, B\}]$ by Sturm sequences. There is none, and the derivative is negative, at both ends of the range of r_1 for every carrier. The derivative is affine in r_1, and the monotonicity part of A10 therefore holds for every r_1 in $[0, 10]$ (E2) and $[0, 2]$ (Proposition 5). The other parts of A10 concern the minimax solution and remain numerical checks.

Use of generative AI

Generative AI tools were used to assist with drafting and editing parts of the text, writing and debugging the simulation, solver and testbed code, and checking bibliographic metadata. The authors designed the study, verified every formal statement and proof, re-ran the artifact that produces all reported numbers, checked every reference against its publisher or archive record, and take full responsibility for the content.